Rohan Keshav Harish

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Early-career Data Scientist and Software Engineer with hands-on experience developing Generative AI systems, ML pipelines, and RAG-based LLM applications. Proven ability to take AI/ML projects from conceptualization to production across the full lifecycle — including data engineering, model evaluation, and agentic workflow deployment.

Education

University of Texas at Arlington — Arlington, TX
Masters of Science in Computer Science - GPA: 3.92 — *Aug. 2023 – May 2025*

Experience

Software Engineer — September 2025 – Present
Arcadis (City of Houston) — *Texas, USA*

- Engineered event-driven ETL automation using AWS Lambda and Python, processing 500K records daily and reducing processing time by 18%, delivering clean, analytics-ready datasets for downstream ML and BI consumption.
- Refactored legacy backend codebase into a parallel computing pipeline using Python Multiprocessing, reducing execution time by 33% while maintaining memory footprint, significantly improving system throughput and backend responsiveness for end users.

AI Engineer — July 2025 – September 2025
University of Texas at Arlington — *Texas, USA*

- Designed and implemented multi-step Agentic AI pipelines in Python using LLMs, translating complex research ideas into working code to automate decision-making workflows and reduce manual iteration cycles by 1.5x.
- Integrated AI agents with digital signal processing, building adaptive systems that dynamically improved contextual signal interpretation.
- Documented and shared research findings with cross-functional collaborators, fostering knowledge-sharing and contributing to team knowledge accumulation.

Data Science Intern — Jul. 2021 – Aug. 2021
The Sparks Foundation

- Architected a supervised ML prediction system, evaluating Linear Regression and Clustering models through rigorous comparative analysis, achieving 89% accuracy on student exam outcomes, demonstrating end-to-end ML model development and evaluation.
- Developed end-to-end MySQL ETL pipelines and drove a 16% precision improvement through systematic feature engineering and exploratory data analysis, building a data-driven foundation for model development.

Projects

Agentic Resume Optimizer | *Python, OpenAI API, ReAct Agent, RAG, SQLite*

- Architected an autonomous AI platform with a ReAct agent loop and 14 OpenAI function-calling tools, enabling end-to-end JD analysis,gap detection, and ATS-optimized output across multiple sessions,demonstrating productive translation of a complex research idea into a fully working, production-grade codebase.
- Engineered a per-user RAG knowledge base with SQLite vector storage and OpenAI embeddings achieving high JD alignment scores, built a template-preserving LaTeX pipeline with brace-aware structural repair and automated one-page enforcement.

Quant Vision | *Python, yfinance, Plotly, Quant Analysis, streamlit, API, Git, statistics, time-series analysis*

- Built an end-to-end data science platform in Python applying statistical modelling, time-series forecasting, and backtesting across a complex quantitative domain, delivering real-time market analytics and risk management insights.
- Elevated investment decision quality using Pandas, NumPy, and statistical modeling to generate risk management insights that improved investment timing accuracy

gRPC File & Compute Server | *gRPC, Multithreading, RPC, Python, Distributed system*

- Developed a multi-threaded gRPC server in Python, supporting distributed file operations and compute through sync/async RPCs.
- Automated file change detection and remote compute execution by deploying a Python folder-watcher client with multithreading, eliminating manual monitoring overhead and enabling real-time triggered workflows.

Megacosm - Detection and Classification of Astronomical Objects | *YOLO v5, CNN, Git, NumPy, OpenCV*

- Engineered a computer vision pipeline in Python with YOLO v5 and OpenCV to detect and classify astronomical objects in real time.
- Boosted model performance by 25% over baseline through iterative CNN optimizations, hyperparameter tuning, and rigorous benchmarking.

Technical Skills

Languages and Frameworks: C++, Python, Java, JavaScript, PHP, HTML, CSS, Typescript, PySpark
Database and Cloud Platforms: SQL, Cassandra, SparkSQL, AWS Lambda, Glue, Redshift, Distributed Systems, Snowflake
Machine Learning and Deep Learning: OpenCV, NLP, Neural Networks, Pandas, Scikit-learn, Deep Learning (GPT)
Tools and Platforms: VS Code, Android Studio, Wireshark, Cisco Packet Tracer, PowerBI, MS, Jira, Palantir foundry, REST API, Docker
AI and Automation: Agentic AI, Workflows, Automations, n8n, RAG, Langchain, Prompt Engineering
UI and Frontend Development: React.js, TailwindCSS, Streamlit, Responsive Design, API Integration

Achievements and Certifications

Publications:MegaCosm – Detection and Classification of Astronomical Objects, IRJCS, Link
Certifications: AWS Cloud Essentials, Google IT Support Professional Certificate, Meta front-end Developer Professional certificate.
Leadership: Event Lead at ESC UTA, Critical thinking, Detail Oriented
Awards: Roll of Honor (Best student of the department), Maverick Distinction, Maverick Advantage